

Problem Statement for Estimated Time of Arrival

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The primary objective of this thesis is to develop a deep learning system, trained over carefully-derived synthetic data, being capable of accurately predicting the estimated time of arrival (ETA) for various urban transportation scenarios. In today's world, accurate ETA predictions are crucial for navigation systems, directly impacting users' planning and decision-making processes. With the ever-increasing number of vehicles on the road and the growing complexity of traffic patterns, such systems are integral for reducing travel time, minimizing fuel consumption, and improving overall efficiency of transportation networks.

By leveraging the Simulation of Urban MObility (SUMO) simulator, extensive driving and traffic simulation data will be gathered. The choice of SUMO is driven by its capability to simulate realistic traffic scenarios and provide detailed, customizable datasets, which will serve as the foundation for training and testing the machine learning models. The data will include simulations of cars, buses, and motorbikes navigating through realistic traffic conditions in a specified real-world area of a map imported into SUMO. These simulations will generate ETA data based on starting and destination points within the mapped area.

In developing the machine learning models, this thesis will utilize Python and TensorFlow due to their powerful libraries and frameworks for deep learning. Various deep learning techniques, such as recurrent neural networks (RNNs) like long short-term memory (LSTMs), will be explored to improve the accuracy of ETA predictions. Data preprocessing and feature engineering will be crucial steps to ensure the quality and relevance of the input data.

To provide context and support for the proposed methods, several key resources and papers on ETA prediction and deep learning in transportation will be referenced:

1. Antypas, E., Spanos, G., Lalas, A., Votis, K., & Tzovaras, D. (2024). A time-series approach for estimated time of arrival prediction in autonomous vehicles.
2. Al-Naim, R., & Lytkin, Y. (2021). Review and comparison of prediction algorithms for the estimated time of arrival using geospatial transportation data.
3. Wang, D., Zhang, J., Cao, W., Li, J., & Zheng, Y. (2018). When Will You Arrive? Estimating Travel Time Based on Deep Neural Networks.
4. Zheng Wang, Kun Fu, & Jieping Ye. (2018). Learning to Estimate the Travel Time.